NLP 201 Entry Quiz

So we know what you know. Please do this by yourself and not with a friend.

Name:

1. Determine whether each statement is TRUE or FALSE, when A, B and C can be any sets. Justify your answer. (Note: To prove set equality $A = B$, you can prove that every element of A is in B and every element of B is in A)

(a) If $A \cup C = B \cup C$, then $A = B$.

(b) If $A \cap C = B \cap C$, then $A = B$.

(c) If $A \cap B = A \cup B$, then $A = B$.

2. Let $A = \{a, b, c\}$. Mark each statement TRUE or FALSE. No need to explain.

(a) $\{b, c\} \in \mathcal{P}(A)$

(b) $\{\{a\}\} \subseteq \mathcal{P}(A)$

(c) $\emptyset \subseteq A$

(d) $\{\emptyset\} \subseteq \mathcal{P}(A)$

(e) $\emptyset \subseteq A \times A$

(f) $\{a, c\} \in A$

(g) $\{a, b\} \in A \times A$

(h) $(c, c) \in A \times A$

3. A club with 20 women and 17 men needs to choose three different members to be president, vice president, and treasurer.

(a) In how many ways is this possible?

(b) In how many ways is this possible if women will be chosen as president and vice president and a man as treasurer?

4. What is the probability that a fair coin lands Heads 3 times out of 5 flips?

5. You pick a word at random from the set of all words of length six of letters of the alphabet with no repeated letters. What is the probability that the word has exactly one vowel?

6. What level is your calculus preparation? Are you familiar with multi-variable derivatives, and/or integrals?

7. Are you comfortable with linear algebra, such as matrix multiplication, eigenvectors, and eigenvalues?

(More on next page.)

8. How much do you know about Python?

9. Anything else you think we might want to know about your background?